FRANCISCO JAVIER CHECA CASAS



Double Degree in Computer Engineering and Business Administration (ADE)

+34 697 92 74 42 • javicheca2003@gmail.com • github.com/javiercheca10 • javiercheca10.github.io • Granada, Spain

EDUCATION

Double Degree in Computer Engineering and Business Administration (ADE)

University of Granada (UGR)

2021 – Present • Granada, Spain

- Comprehensive 5-year curriculum integrating computer science, software engineering, corporate finance, strategic management, and econometrics.
- Specialization in advanced algorithms, distributed systems, operations optimization, and quantitative business modeling.

Erasmus+ International Academic Exchange (Computer Engineering)

University of Piraeus (Department of Informatics)

2024 – 2025 • Athens, Greece

- Specialization in Internet technologies, client-server web architectures (PHP, MySQL), and computer networks in a bilingual international environment.

SICUE National Mobility Exchange (Business Administration)

Universitat Politècnica de València (UPV)

Valencia, Spain

- Coursework focused on strategic management, commercial operations, financial modeling, and corporate decision-making systems.

PROJECTS

RepartoPersonal | Combinatorial Shift & Staff Optimization System

github.com/javiercheca10/RepartoPersonal

- Developed a mathematical optimization engine based on constraint satisfaction programming (CP-SAT) in 3 hierarchical stages, guaranteeing 100% coverage in mandatory posts and eliminating labour compliance violations through hard constraints on skills, vetos, and rest periods.
- Implemented a minimax balancing algorithm for equitable workload distribution and ergonomic rotation, achieving an 85% reduction in fatigue point dispersion and preventing employees from repeating demanding posts for more than 2 consecutive days.
- Engineered a decoupled architecture with a PySide6 (Qt) GUI and autonomous export modules, automating official multi-sheet Excel rosters and executive vector PDFs (ReportLab) in under 5 seconds of computation.
- Built an automated test suite across data pipelines and solver stages, validating edge cases such as staffing shortages, unexpected sick leave, and shift calendar compatibility.

SUMMARY

Double Degree student in Computer Science and Business Administration (UGR) with international academic exchange in Athens (University of Piraeus) and Valencia (UPV). Hybrid profile focused on Software Engineering (Full Stack / Backend), Algorithmic Problem Solving, and Operations Research / Mathematical Optimization applied to business operations. Demonstrated technical expertise in designing constraint programming models with Google OR-Tools CP-SAT, asynchronous APIs and Docker containers with FastAPI, high-performance C++ software, and modern web applications with TypeScript and Astro. Strong analytical mindset capable of translating complex business requirements into scalable, testable software architectures.

SKILLS

Programming Languages

Python | C++ | Java | TypeScript | JavaScript (ES6+) | C | SQL | PHP | Bash / Shell Scripting

Frameworks & Libraries

FastAPI | Astro | PySide6 / Qt | Express.js | Node.js | ReportLab | OpenPyXL | Tailwind CSS | HTML5 | CSS3

Tools & DevOps

Docker | Docker Compose | Git | GitHub Actions (CI/CD) | Linux (Arch, Ubuntu) | Valgrind | Make / CMake | uv | npm | Uvicorn

PROJECTS (CONTINUED)

Torrent Stream Web | Self-Hosted VOD Media Server & Microservices

github.com/javiercheca10/Torrent_stream_web

- Designed and deployed a distributed microservices architecture using Docker Compose (Web API, interactive bot, and sequential BitTorrent engine), achieving sub-15s playback start latency via a two-phase predictive buffering mechanism (512 MB initial buffer and up to 50 GB background prefetch).
- Developed an asynchronous FastAPI backend supporting partial content streaming (HTTP Range Requests), enabling seamless seek, pause, and rewind functionality on Smart TVs, Kodi, and VLC across local networks without bandwidth saturation.
- Implemented an automated lifecycle management subsystem with TTL-based garbage collection (session and inactivity timers) and disk-pressure protection, autonomously purging 100% of temporary files upon reaching 60 GB usage or a 5 GB free disk margin.
- Hardened system security with ephemeral signed session tokens, Telegram Bot API user whitelisting, and rootless Docker container execution under least privilege (cap_drop ALL, read-only filesystem).

Autonomous Deliberative Agent | Artificial Intelligence & Heuristic Search

github.com/javiercheca10/Practica-2-IA

- Implemented a deliberative intelligent agent in C++ for navigation and survival in dynamic grid environments with partial observability, optimizing decision-making and reducing battery consumption by over 40% compared to uninformed reactive strategies.
- Formulated and calibrated admissible and consistent heuristics for the A* search algorithm, ensuring mathematical optimality on state-space graphs with heterogeneous friction matrices (water, forest, cliffs, and roads).
- Designed compact data structures and efficient memory management in C++ for sensory state representation and dynamic inventory handling, achieving sub-millisecond per-cycle decision latencies and collision avoidance in multi-agent settings.

SKILLS (CONTINUED)

Databases & Cloud

MySQL · PostgreSQL · MongoDB · SQLite · Google OR-Tools (CP-SAT) · Nginx · RESTful APIs · HTTP Range Streaming

Concepts & Methodologies

Constraint Programming (CSP) · Heuristic Search (A*, BFS, Dijkstra) · Combinatorial Optimization · Object-Oriented Programming (OOP) · MVC Pattern · Microservices · Container Security · Agile / Scrum · Operations Management · Financial Modeling

LANGUAGES

Spanish

Native

Mother tongue

English

Professional Working Proficiency • B2 First Certified

Cambridge Assessment English • C1 operational

French

Basic (A2)

Elementary proficiency